In [7]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix
from sklearn.preprocessing import StandardScaler
```

In [8]:
```
df = pd.read_csv("C:/Users/mestr/Downloads/archive (11)/diabetes.csv")

df.head()
```

Out[8]:

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	Pedigree	Age	O
0	6	148	72	35	0	33.6	0.627	50	
1	1	85	66	29	0	26.6	0.351	31	
2	8	183	64	0	0	23.3	0.672	32	
3	1	89	66	23	94	28.1	0.167	21	
4	0	137	40	35	168	43.1	2.288	33	

In [9]:
```
X = df.drop("Outcome", axis=1)
y = df["Outcome"]
```

In [10]:
```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

In [11]:
```
sc = StandardScaler()

X_train = sc.fit_transform(X_train)
X_test = sc.transform(X_test)
```

In [12]:
```
sc = StandardScaler()

X_train = sc.fit_transform(X_train)
X_test = sc.transform(X_test)
```

In [14]:
```
knn = KNeighborsClassifier(n_neighbors=5)

knn.fit(X_train, y_train)

y_pred = knn.predict(X_test)
```

In [15]:
```
cm = confusion_matrix(y_test, y_pred)
print("Confusion Matrix:")
print(cm)
```

```
Confusion Matrix:
[[79 20]
 [27 28]]
```

In [18]:
```
cm = confusion_matrix(y_test, y_pred)
acc = accuracy_score(y_test, y_pred)

print("Confusion Matrix:\n", cm)
print("Accuracy:", acc)
print("Error Rate:", 1-acc)
print("Precision:", precision_score(y_test, y_pred))
print("Recall:", recall_score(y_test, y_pred))
```

```
Confusion Matrix:
 [[79 20]
 [27 28]]
Accuracy: 0.6948051948051948
Error Rate: 0.30519480519480524
Precision: 0.5833333333333334
Recall: 0.509090909090909
```

In [ ]: